MATH 152
Mrs. Bonny Tighe

EXAM IA

6.1-7.5
50 points
PART TWO

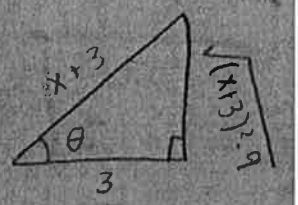
NAME Solutions - Partial Credit

TA_____ Wed10/6/2021

1. (10 points) Integrate $\int \frac{(x+3)^2}{\sqrt{x^2+6x}}\,dx$

$x^2+6x+9-9$

$(x+3)^2-9$



$\int \frac{(x+3)^2}{\sqrt{(x+3)^2-9}}\,dx$

$x+3=3\sec\theta \qquad \sec\theta=\frac{x+3}{3}$

$\frac{dx}{d\theta}=3\sec\theta\tan\theta$

$dx=3\sec\theta\tan\theta\,d\theta$

$\int \frac{9\sec^2\theta}{\sqrt{9\sec^2\theta-9}}\,3\sec\theta\tan\theta\,d\theta$

$u=\sec\theta$
$du=\sec\theta\tan\theta\,d\theta$
$dv=\sec^2\theta\,d\theta$
$v=\tan\theta$

$\int \frac{9\sec^2\theta}{\sqrt{\tan^2\theta}}\,\sec\theta\tan\theta\,d\theta$

$\int \frac{9\sec^2\theta}{\tan\theta}\,\sec\theta\tan\theta\,d\theta$

$9\int\sec^3\theta\,d\theta$

$9\int\sec\theta\sec^2\theta\,d\theta$

$9\left(\sec\theta\tan\theta-\int\sec\theta\tan^2\theta\,d\theta\right)$

$9\left(\sec\theta\tan\theta-\int -\sec\theta+\sec^3\theta\,d\theta\right)$

$9\sec\theta\tan\theta+9\int\sec\theta\,d\theta-9\int\sec^3\theta\,d\theta$

$\frac{1}{18}\left(9\sec\theta\tan\theta+9\ln|\sec\theta+\tan\theta|+C\right)$

$\frac{1}{18}\left(9\left(\frac{x+3}{3}\right)\left(\frac{\sqrt{(x+3)^2-9}}{3}\right)+9\ln\left|\frac{x+3}{3}+\frac{\sqrt{(x+3)^2-9}}{3}\right|\right)+C$

$\boxed{\frac{1}{2}(x+3)\left(\sqrt{(x+3)^2-9}\right)-\frac{1}{2}\ln\left|x+3+\sqrt{(x+3)^2-9}\right|+C}$

$\int \to \frac{a}{n}=\frac{\sqrt{(x+3)^2-9}}{x+3}$

$c \to \frac{a}{n}=\frac{3}{x+3}$

$t \to \frac{a}{a}=\frac{\sqrt{(x+3)^2-9}}{3}$

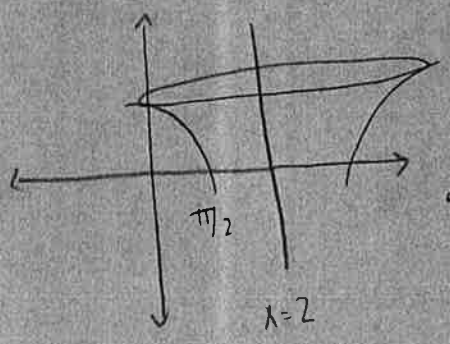
$\sec \to \frac{n}{a}=\frac{x+3}{3}$

$\csc \to \frac{n}{o}=\frac{x+3}{\sqrt{(x+3)^2-9}}$

$\cot \to \frac{a}{o}=\frac{3}{\sqrt{(x+3)^2-9}}$

2. (10 points) Find the volume of the solid of revolution for the region bounded by the curves $y=\cos x$ *and the* $x-axis$ on the interval $0\le x\le\frac{\pi}{2}$ rotated about the line x = 2.

Graph



$2\pi\int_0^{\pi/2}(2-x)(\cos x)\,dx$

$u=x \qquad dv=\cos x\,dx$
$du=dx \qquad v=\sin x$

$\int_0^{\pi/2}2\cos x\,dx-\int_0^{\pi/2}x\cos x\,dx$

$2\sin x-x\sin x+\int_0^{\pi/2}\sin x\,dx$

$2\sin x-x\sin x-\cos x\ \Big|_0^{\pi/2}$

$\left(2\sin(\pi/2)-(\pi/2)\sin(\pi/2)-\cos(\pi/2)\right)-\left(2\sin(0)-0-\cos(0)\right)$

$2-\pi/2+1$

$2\pi\left(\boxed{3-\pi/2}\right)$

3. (10 points) Integrate: $\int \sin 3t\, e^{2t}\, dt$

$u = \sin 3t \qquad dv = e^{2t}\,dt$

$du = 3\cos 3t\,dt \qquad v = \frac{e^{2t}}{2}$

1

$\frac{1}{2}e^{2t}\sin 3t - \int \frac{1}{2}e^{2t}\,3\cos 3t\,dt$ 2

$\frac{1}{2}e^{2t}\sin 3t - \frac{3}{2}\int e^{2t}\cos 3t\,dt$

$u = \cos 3t \qquad dv = e^{2t}dt$

$du = -3\sin 3t\,dt \qquad v = \frac{e^{2t}}{2}$

$-\frac{3}{2}\left[\frac{e^{2t}}{2}\cos 3t - \int \frac{e^{2t}}{2}(-3)\sin 3t\,dt\right.$

2

$= \frac{1}{2}e^{2t}\sin 3t - \frac{3}{4}e^{2t}\cos 3t + \frac{3}{2}\int -\frac{3}{2}e^{2t}\sin 3t\,dt$

$\frac{4}{4}\text{☺} = \frac{1}{2}e^{2t}\sin 3t - \frac{3}{4}e^{2t}\cos 3t - \frac{9}{4}\int e^{2t}\sin 3t\,dt$

$+\frac{9}{4}\text{☺}$

$+\frac{9}{4}\text{☺}$ 1

$\frac{13}{4}\text{☺} = \frac{1}{2}e^{2t}\sin 3t - \frac{3}{4}e^{2t}\cos 3t + C$

$\text{☺} = \frac{4}{13}\left(\frac{1}{2}e^{2t}\sin 3t - \frac{3}{4}e^{2t}\cos 3t\right) + C = \boxed{\frac{2}{13}e^{2t}\sin 3t - \frac{3}{13}e^{2t}\cos 3t + C}$ 2

$u = e^{2t} \qquad dv = \sin 3t\,dt$

$du = 2e^{2t}\,dt \qquad v = -\frac{1}{3}\cos 3t$ 1

$e^{2t}\left(-\frac{1}{3}\cos 3t\right) - \int\left(-\frac{1}{3}\cos 3t\right)(2e^{2t}\,dt)$ 2

$-\frac{1}{3}e^{2t}\cos 3t + \frac{2}{3}\int \cos 3t\, e^{2t}\,dt$

$u = e^{2t} \qquad dv = \cos 3t\,dt$

$du = 2e^{2t}\,dt \qquad v = \frac{\sin 3t}{3}$

2

$\frac{2}{3}\left[(e^{2t})\left(\frac{\sin 3t}{3}\right) - \int \frac{\sin 3t}{3}\,2e^{2t}\,dt\right.$

2

$\frac{9}{9}\text{☺} = -\frac{1}{3}e^{2t}\cos 3t + \frac{2}{9}e^{2t}\sin 3t - \frac{4}{9}\int \sin 3t\, e^{2t}\,dt$

$+\frac{4}{9}\text{☺}$

$+\frac{4}{9}\text{☺}$ 1

$\frac{13}{9}\text{☺} = -\frac{1}{3}e^{2t}\cos 3t + \frac{2}{9}e^{2t}\sin 3t + C$

$\text{☺} = \frac{9}{13}\left(-\frac{1}{3}e^{2t}\cos 3t + \frac{2}{9}e^{2t}\sin 3t\right) + C$

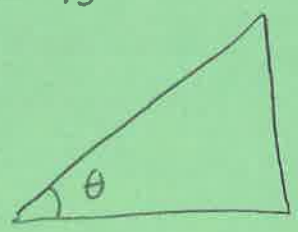
4. (10 points) Integrate: $\int_3^6 \frac{x^3}{\sqrt{x^2-9}}\,dx$

$x = \sec\theta$

$dx = 3\sec\theta\tan\theta\,d\theta$ 1

$\frac{x}{3} = \sec\theta = \frac{3}{3} = 1 \qquad \theta = 0$

$\sec\theta = \frac{6}{3} = 2 \qquad \theta = \pi/3$



$\int_0^{\pi/3} \frac{27\sec^3\theta}{\sqrt{9\sec^2\theta - 9}}\,3\sec\theta\tan\theta\,d\theta$ 1

$\int_0^{\pi/3} \frac{27\sec^3\theta}{\sqrt{9\tan^2\theta}}\,3\sec\theta\tan\theta\,d\theta$

$\int_0^{\pi/3} \frac{27\sec^3\theta}{3\tan\theta}\,3\sec\theta\tan\theta\,d\theta$ 1

$27\int_0^{\pi/3} \sec^4\theta\,d\theta$ 1

$27\int_0^{\pi/3} \sec^2\theta(1+\tan^2\theta)\,d\theta$ 1

$u = \tan\theta$

$du/d\theta = \sec^2\theta$

$d\theta = du/\sec^2\theta$

$27\int 1 + u^2\,du$ 1

$27\left[u + \frac{1}{3}u^3\right]$ 1

$27\left(\tan\theta + \frac{1}{3}\tan^3\theta\right)\Big|_0^{\pi/3}$ 1

$27\left(\tan(\pi/3) + \frac{1}{3}(\tan\pi/3)^3\right) - 27\left(\tan(0) + \frac{1}{3}\tan(0)^3\right)$

$27\left(\sqrt{3} + \frac{1}{3}\sqrt{3}^3\right) = 27(\sqrt{3} + \sqrt{3}) = 27(2\sqrt{3}) = \boxed{54\sqrt{3}}$ 1

→ *could change bounds here to u

$u = \tan\theta$

$u = \tan(\pi/3) = \sqrt{3}$

$u = \tan(0) = 0$

*note→

$(\sqrt{3})^3 = \sqrt{3}\cdot\sqrt{3}\cdot\sqrt{3}$

$= 3\sqrt{3}$

First: divide!

$$\begin{array}{r} x \\ x^4+x^2 \overline{\big) x^5+2} \\ -(x^5+x^3) \\ \hline -x^3+2 \end{array}$$

5. (10 points) Integrate $\int \frac{x^5+2}{x^4+x^2}\,dx$

+1 $\int x + \frac{-x^3+2}{x^2(x^2+1)}\,dx$

$\frac{-x^3+2}{x^2(x^2+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+1}$

$A(x)(x^2+1) + B(x^2+1) + (Cx+D)(x^2) = -x^3+2$

$Ax^3 + Ax + Bx^2 + B + Cx^3 + Dx^2 = -x^3+2$

$(A+C)x^3 + (B+D)x^2 + Ax + B = -x^3+2$

$A+C=-1$ $\quad B+D=0$ $\quad \boxed{A=0}$ $\quad \boxed{B=2}$

$0+C=-1$ $\quad 2+D=0$

$\boxed{C=-1}$ $\quad \boxed{D=-2}$

+4

1 $\int x + \frac{2}{x^2} + \frac{-x-2}{x^2+1}\,dx$

2 $\int x + \frac{2}{x^2} - \frac{x}{x^2+1} - \frac{2}{x^2+1}\,dx$

$u = x^2+1$

$du/dx = 2x$

$dx = du/2x$

$\int x\,dx + \int \frac{2}{x^2}dx - \int \frac{1}{2}\left(\frac{1}{u}\right)du - \int \frac{2}{x^2+1}\,dx$

$\boxed{\frac{x^2}{2} - \frac{2}{x} - \frac{1}{2}\ln|x^2+1| - 2\arctan x + C}$

+1

6. (5 points) Integrate $\int \frac{1}{3^{2x}+1}\,dx$

$m = 3^x$

$\frac{dm}{dx} = 3^x \ln 3$

$dx = dm/m\ln 3$

$\int \frac{1}{(m^2+1)} \frac{dm}{m\ln 3}$

$\frac{1}{\ln 3}\int \frac{1}{m(m^2+1)}\,dm \longrightarrow \frac{1}{m(m^2+1)} = \frac{A}{m} + \frac{Bm+C}{m^2+1}$

$A(m^2+1) + (Bm+C)m = 1$

$Am^2 + A + Bm^2 + Cm = 1$

$(A+B)m^2 + Cm + A = 1$

$\boxed{C=0}$ $\quad \boxed{A=1}$

$A+B=0$

$1+B=0$

$\boxed{B=-1}$

$\frac{1}{\ln 3}\int \frac{1}{m} - \frac{m}{m^2+1}\,dm$

$u = m^2+1$

$du/dm = 2m$

$dm = du/2m$

$\frac{1}{\ln 3}\left[\ln|m| - \frac{1}{2}\int \frac{1}{u}\,du\right]$

$\frac{1}{\ln 3}\ln|m| - \frac{1}{2\ln 3}\ln|m^2+1| + C$

$\boxed{\frac{1}{\ln 3}\ln|3^x| - \frac{1}{2\ln 3}\ln|3^{2x}+1| + C}$

MATH 152 **EXAM I** NAME Solutions - partial credit

Mrs. Bonny Tighe 6.1-7.5

$4\int\sec^3\theta = 4\sec\theta\tan\theta - 4\int\sec^3\theta\, d\theta$ 50 points TA_____ Wed10/6/2021

$-4\ln|\sec\theta+\tan\theta|$ **PART TWO**

$\int\sec^3\theta = \frac{1}{2}\sec\theta\tan\theta - \frac{1}{2}\ln|\sec\theta+\tan\theta|$

1. (10 points) Integrate $\int\frac{(x+2)^2}{\sqrt{x^2+4x+8}}dx$

$x^2+4x+8+4-4$

$(x+2)^2+4$

$x+2 = 2\tan\theta$

$dx = 2\sec^2\theta\, d\theta$

$\int\frac{4\tan^2\theta}{\sqrt{4\tan^2\theta+4}}(2\sec^2\theta\, d\theta)$

$\int\frac{4\tan^2\theta}{\sqrt{4(\sec^2\theta)}}(2\sec^2\theta\, d\theta)$

$\int\frac{4\tan^2\theta}{\cancel{2\sec\theta}}\cancel{2}\sec^2\theta\, d\theta$

$s \to h/a$

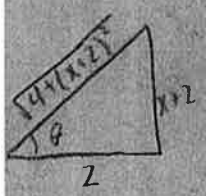
$t \to o/a$

$4\int\tan^2\theta\sec\theta\, d\theta$

$4\int(\sec^2\theta-1)(\sec\theta)\, d\theta$

$4\int\sec^3\theta - \sec\theta\, d\theta$

$4\int\sec^3\theta\, d\theta - 4\int\sec\theta\, d\theta$



$u = \sec\theta$ $dv = \sec^2\theta$

$du = \sec\theta\tan\theta\, d\theta$ $v = \tan\theta$

$4(\sec\theta\tan\theta - \int\sec\theta\tan^2\theta\, d\theta) - 4\int\sec\theta\, d\theta$

$4\sec\theta\tan\theta - 4\int\sec\theta(\sec^2\theta-1)\, d\theta - 4\int\sec\theta\, d\theta$

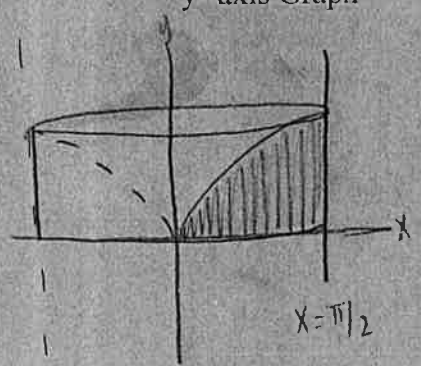
$4\sec\theta\tan\theta - 4\int\sec^3\theta\, d\theta - 4\int\sec\theta\, d\theta - 4\int\sec\theta\, d\theta$

$\frac{1}{2}\sec\theta\tan\theta - \frac{1}{2}\ln|\sec\theta+\tan\theta| - 4\ln|\sec\theta+\tan\theta|$

$\frac{1}{2}\sec\theta\tan\theta - \frac{9}{2}\ln|\sec\theta+\tan\theta| + C$

$\frac{1}{2}\left(\frac{\sqrt{x^2+4x+8}}{2}\right)\left(\frac{x+2}{2}\right) - \frac{9}{2}\left(\frac{\sqrt{x^2+4x+8}}{2} + \frac{x+2}{2}\right) + C$

2. (10 points) Find the volume of the solid of revolution for the region bounded by the curves $y = \sin x$ *and the* $x-$*axis* on the interval $0 \le x \le \frac{\pi}{2}$ rotated about the y- axis Graph



$2\pi\int_a^b r\cdot h$

$u = x$ $dv = \sin x\, dx$

$du = dx$ $v = -\cos x$

$2\pi\int_0^{\pi/2} x\sin x\, dx$

$2\pi\left(-x\cos x\Big|_0^{\pi/2} - \int_0^{\pi/2} -\cos x\, dx\right)$

$2\pi\left(-x\cos x\Big|_0^{\pi/2} + \int_0^{\pi/2}\cos x\, dx\right)$

$2\pi\left(-x\cos x + \sin x\Big|_0^{\pi/2}\right)$

$2\pi\left(-\pi/2\cos(\pi/2) + \sin(\pi/2) - (-(0)\cos(0) + \sin(0))\right)$

$2\pi\left(-\pi/2(0) + (1) - (-0+0)\right)$

$\boxed{2\pi}$

3. (10 points) Integrate: $\int \cos 4t\, e^{3t}\, dt$

1pt $\boxed{u = \cos 4t \quad dv = e^{3t}\,dt \qquad du = -4\sin 4t\,dt \quad v = \frac{1}{3}e^{3t}}$

2 pts

$= \boxed{\frac{1}{3}e^{3t}\cos 4t - \int \frac{1}{3}e^{3t}(-4\sin 4t)\,dt}$

1 pt

$= \frac{1}{3}e^{3t}\cos 4t + \frac{4}{3}\int e^{3t}\sin 4t\,dt$

$\boxed{u = \sin 4t \quad dv = e^{3t}dt \qquad du = 4\cos 4t\,dt \quad v = \frac{1}{3}e^{3t}}$

$= \frac{1}{3}e^{3t}\cos 4t + \boxed{\frac{4}{3}\left[\frac{1}{3}e^{3t}\sin 4t - \int \frac{1}{3}e^{3t}\,4\cos 4t\,dt\right]}$ 1pt

$= \boxed{\frac{1}{3}e^{3t}\cos 4t + \frac{4}{9}e^{3t}\sin 4t - \frac{16}{9}\int e^{3t}\cos 4t\,dt}$ 2 pts

$+\frac{16}{9}$ ☺ $\boxed{+\frac{16}{9}\text{ ☺}}$ 1 pt

$25/9\int \cos 4t\, e^{3t}\,dt = \frac{1}{3}e^{3t}\cos 4t + \frac{4}{9}e^{3t}\sin 4t + C$

$\int \cos 4t\, e^{3t}\,dt = \frac{9}{25}\left[\frac{1}{3}e^{3t}\cos 4t + \frac{4}{9}e^{3t}\sin 4t\right] + C$

or

2 pts

$\boxed{\frac{3}{25}e^{3t}\cos 4t + \frac{4}{25}e^{3t}\sin 4t + C}$

or

$\boxed{u = e^{3t} \quad dv = \cos 4t\,dt \qquad du = 3e^{3t}\,dt \quad v = \frac{1}{4}\sin 4t}$ 1pt

$\boxed{= \frac{1}{4}e^{3t}\sin 4t - \int \frac{1}{4}\sin 4t\, 3e^{3t}\,dt}$ 2 pts

$\boxed{dv = \sin 4t \quad v = \frac{1}{4}\cos 4t}$

$= \frac{1}{4}e^{3t}\sin 4t - \frac{3}{4}\int \sin 4t\, e^{3t}\,dt$

$\boxed{u = e^{3t} \quad du = 3e^{3t}\,dt}$ 1pt

$= \frac{1}{4}e^{3t}\sin 4t - \boxed{\frac{3}{4}\left[\frac{1}{4}e^{3t}\cos 4t - \int \frac{1}{4}\cos 4t\, 3e^{3t}\,dt\right]}$ 1pt

$\boxed{= \frac{1}{4}e^{3t}\sin 4t + \frac{3}{16}e^{3t}\cos 4t - \frac{9}{16}\int \cos 4t\, e^{3t}\,dt}$ 2 pts

$+\frac{9}{16}$ ☺ $\boxed{+\frac{9}{16}\text{ ☺}}$ 1pt

$25/16\int \cos 4t\, e^{3t}\,dt = \frac{1}{4}e^{3t}\sin 4t + \frac{3}{16}e^{3t}\cos 4t + C$

$\int \cos 4t\, e^{3t}\,dt = \frac{16}{25}\left[\frac{1}{4}e^{3t}\sin 4t + \frac{3}{16}e^{3t}\cos 4t\right] + C$

4. (10 points) Integrate: $\int_{\sqrt{3}}^{3} \frac{x^3}{\sqrt{9+x^2}}\,dx$

$\boxed{x = 3\tan\theta \quad dx = 3\sec^2\theta\,d\theta}$ 1pt

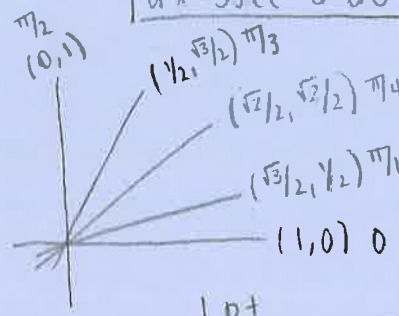
$\tan\theta = x/3$

$\tan\theta = 3/3 = 1$

$\theta = \pi/4$

$\tan\theta = \sqrt{3}/3$

$\theta = \pi/6$



$\boxed{\int_{\pi/6}^{\pi/4} \frac{27\tan^3\theta}{\sqrt{9+9\tan^2\theta}}\,3\sec^2\theta\,d\theta}$ 1pt

$\boxed{27\int_{\pi/6}^{\pi/4} \frac{\tan^3\theta}{\sqrt{9(1+\tan^2\theta)}}\,3\sec^2\theta\,d\theta}$ 1pt

$\boxed{27\int_{\pi/6}^{\pi/4} \frac{\tan^3\theta}{\cancel{3}\sqrt{\sec^2\theta}}\,\cancel{3}\sec^2\theta\,d\theta}$ 1pt

$27\int_{\pi/6}^{\pi/4} \frac{\tan^3\theta}{\sec\theta}\sec^2\theta\,d\theta$

$\boxed{27\int_{\pi/6}^{\pi/4} \tan^3\theta\sec\theta\,d\theta}$ 1pt

$27\int_{\pi/6}^{\pi/4} \tan^2\theta\tan\theta\sec\theta\,d\theta$

$\boxed{27\int_{\pi/6}^{\pi/4} (\sec^2\theta - 1)\tan\theta\sec\theta\,d\theta}$ 1pt

$u = \sec\theta$

$du/d\theta = \sec\theta\tan\theta$

$\boxed{27\int_{\pi/6}^{\pi/4} u^2 - 1\,du}$

1pt

$27\left[\frac{u^3}{3} - u\right]$

you can leave the bounds as x's and there's an extra step step at the end

$\boxed{27\,\frac{\sec^3\theta}{3} - \sec\theta\,\Big|_{\pi/6}^{\pi/4}}$ 1pt

$\boxed{27\left[\frac{1}{3}\sec(\pi/4)^3 - \sec(\pi/4) - \frac{1}{3}\sec^3(\pi/6) + \sec(\pi/6)\right]}$

$27\left[\frac{1}{3}(2/\sqrt{2})^3 - (2/\sqrt{2}) - \frac{1}{3}(2/\sqrt{3})^3 + (2/\sqrt{3})\right]$

$27\left[\frac{1}{3}(8/2\sqrt{2}) - (2/\sqrt{2}) - \frac{1}{3}(8/3\sqrt{3}) + (2/\sqrt{3})\right]$

$27\left[\frac{1}{3}\left(\frac{4}{\sqrt{2}}\right) - \left(\frac{2}{\sqrt{2}}\right) - \frac{1}{3}\left(\frac{8}{3\sqrt{3}}\right) + \frac{2}{\sqrt{3}}\right]$

$27\left[\frac{4}{3\sqrt{2}} - \frac{6}{3\sqrt{2}} - \frac{8}{9\sqrt{3}} + \frac{18}{9\sqrt{3}}\right]$

$27\left[\frac{-2}{3\sqrt{2}} + \frac{10}{9\sqrt{3}}\right]$

$\frac{-18}{\sqrt{2}} + \frac{30}{\sqrt{3}} \Rightarrow \frac{-18\sqrt{2}}{2} + \frac{30\sqrt{3}}{3} \Rightarrow \boxed{-9\sqrt{2} + 10\sqrt{3}}$ 1pt

$$x^4+x^2 \overline{\big) x^5 + \dots + 2} \quad x$$
$$-(x^5+x^3)$$
$$-x^3+2$$

5. (10 points) Integrate $\int \frac{x^5+2}{x^4+x^2}\,dx$

$$\int x + \frac{-x^3+2}{x^2(x^2+1)}\,dx$$

$$\int x + \frac{2}{x^2} + \frac{-x-2}{x^2+1}\,dx$$

$$\int x + 2x^{-2} - \frac{x}{x^2+1} - \frac{2}{x^2+1}\,dx$$

$u = x^2+1$
$du = 2x\,dx$

$$\frac{1}{2}x^2 - \frac{2}{x} - \int \frac{1}{u}\,du - 2\tan^{-1}x + C$$

$$\frac{1}{2}x^2 - \frac{2}{x} - \ln|x^2+1| - 2\tan^{-1}x + C$$

$$\frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+1} = \frac{-x^3+2}{x^2(x^2+1)}$$

$$A(x)(x^2+1) + B(x^2+1) + (Cx+D)(x^2) = -x^3+2$$
$$Ax^3 + Ax + Bx^2 + B + Cx^3 + Dx^2 = -x^3+2$$
$$(A+C)x^3 + (B+D)x^2 + Ax + B = -x^3+2$$

$A+C=-1$ $\boxed{C=-1}$

$B+D=0$ $\boxed{A=0}$ $\boxed{B=2}$

$2+D=0$ $\boxed{D=-2}$

6. (5 points) Integrate $\int \frac{1}{1+5^{2x}}\,dx$

$m = 5^x$ $\quad dm = 5^x \ln 5\,dx$

$m^2 = 5^{2x}$ $\quad dx = \frac{dm}{m\ln 5}$

$$\int \frac{1}{1+m^2}\left(\frac{dm}{m\ln 5}\right)$$

$$\frac{1}{\ln 5}\int \frac{1}{m(1+m^2)}\,dm$$

$$\frac{1}{\ln 5}\int \frac{1}{m} + \frac{-m}{1+m^2}\,dm$$

$u = 1+m^2$
$du = 2m\,dm$

$$\frac{1}{\ln 5}\left[\ln|m| - \frac{1}{2}\ln|1+m^2|\right] + C$$

$$\frac{1}{\ln 5}\left(\ln|5^x| - \frac{1}{2}\ln|1+5^{2x}|\right) + C$$

$$\frac{A}{m} + \frac{Bm+C}{1+m^2} = \frac{1}{m(1+m^2)}$$

$$A(1+m^2) + (Bm+C)m = 1$$
$$A + Am^2 + Bm^2 + Cm = 1$$
$$(A+B)m^2 + Cm + A = 1$$

$A+B=0$ $\boxed{A=1}$

$1+B=0$ $\boxed{C=0}$

$\boxed{B=-1}$